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BOLTED END PLATE MOMENT CONNECTIONS

PURPOSE

This document directs users to the appropriate resources for the design of bolted end plate moment connections.

SCOPE

This document includes information about the design of end plate beam-to-column flange moment connections with and without seismic detailing requirements. Moment connections may be used on pipe racks, equipment structures, and buildings.

APPLICATION

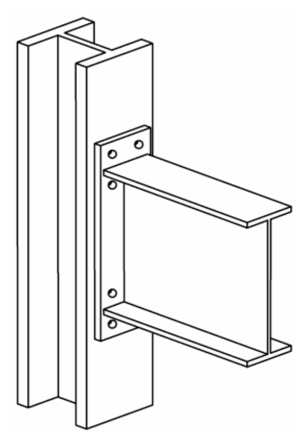
This guideline may not be based on the current codes, specifications and standards. Prior to use on a project, the user shall verify compliance with codes, specifications and standards in effect for the project.

Connection Stiffness

Bolt end-plate moment connections are intended to perform as fully-restrained (FR) moment connections. AISC 360 states: "A fully-restrained (FR) moment connection transfers moment with a negligible rotation between the connected members. In the analysis of the structure, the connection may be assumed to allow no relative rotation."

Four Bolt Unstiffened (4E)

Unstiffened 4 tension bolt connections (8 total bolts) are easy to fabricate and erect, and are the most economical type of moment connection. For large moments, use of a deeper beam or deviation from standard AISC horizontal gage will help in the design of the standard 4 tension bolt connection.



Refer to standard drawing 000.215.5040 for connection geometry where no seismic detailing is required. Refer to standard drawing 000.215.5065 for connection geometry where seismic detailing is required.

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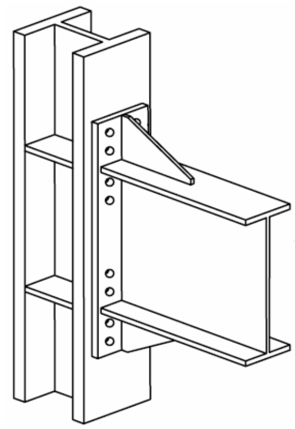
Eight Bolt Stiffened (8ES)

Stiffened 8 tension bolt connections (16 total bolts) will carry larger moments than the standard 4 tension bolt connection. This connection requires stiffeners above the top flange and below the bottom flange at the end plate that may reduce available space above the connection for piping, access, and other similar functions.



Refer to standard drawing 000.215.5040 for connection geometry where no seismic detailing is required. Refer to standard drawing 000.215.5065 for connection geometry where seismic detailing is required.

SEISMIC DETAILING

General

Moment connections are designed for gravity and lateral loads. When seismic detailing is required by ASCE 7, designs must be in accordance with AISC 341. For all other situations, design procedures prescribed in AISC Design Guide 4 are suitable.

Seismic Detailing Required

Seismic detailing in accordance with AISC 341 may be required by either ASCE 7 Table 12.2-1 or ASCE 7 Table 15.4-1. AISC 341 is a supplement to AISC 360 and includes requirements for Ordinary Moment Frames (OMF), Intermediate moment Frames (IMF), and Special Moment Frames (SMF). AISC 358 is a supplement to AISC 341 that specifically addresses prequalified beam-to-column connections for IMFs and SMFs.

AISC 358 prescribes design procedures for bolted end-plate moment connections for use in IMFs and SMFs. AISC 341 permits the use of these procedures for OMFs also. As a result, it is recommended that AISC 358 design procedures be used whenever seismic detailing is required by ASCE 7.

Note!!! It is appropriate to use engineering judgment when exceeding the parametric limitations of AISC 358 Table 6.1 for rolled shapes used in OMFs. The commentary indicates that "There is no evidence that modest deviations from these dimensions will result in significant different performance."

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When seismic detailing is required, the design procedure should include the following:

- Select moment frame system and corresponding R value from either ASCE 7 Table 12.2-1 or ASCE 7 Table 15.4-1. Guideline 000.215.1216 includes a discussion regarding consideration of SMF, IMF or OMF systems.
- Select member sizes in accordance with AISC 360.
- Design bolted end-plate moment connections in accordance with AISC 358.
- Complete other detailing requirements in accordance with AISC 341.

See Attachment 01 for a sample design.

No Seismic Detailing Required

When seismic detailing is not required by either ASCE 7 Table 12.2-1 or ASCE 7 Table 15.4-1, the design procedure should include the following:

- Select member sizes in accordance with AISC 360.
- Design bolted end-plate moment connections in accordance with AISC Design Guide 4.

See AISC Design Guide 4 for several example problems.

Note!!! AISC Design Guide 16 also gives procedures for bolted end-plate moment connections, where the bolts are within the beam flanges. These configurations are seldom used at Fluor, are not covered by Fluor standard drawings, and are not addressed in this document.

COMPUTER DESIGN

Design of bolted end plate moment connections should be done using electronic spreadsheets or similar computer programs if available. Consult the Lead Structural Engineer for availability and use of computer programs.

Note!!! Commercially available software, such as DesconWin, is a Fluor reference program, and is accessible from MySoftware

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BOLTED END PLATE MOMENT CONNECTIONS

CONNECTION DETAILS

Design and detailing should be in accordance with the appropriate Fluor standard drawings and the appropriate AISC documents.

3/4-inch diameter ASTM A325 bolts will normally be used. Larger diameter bolts, up to 1- 1/2-inch, and A490 bolts may be used for heavily loaded connections. Only A325 bolts may be used for the stiffened 8 tension bolt connection.

Width of end plates should be approximately 1-inch larger than the beam flange width, and should not exceed column flange width.

REFERENCES

AISC 341-05, Seismic Provisions For Structural Steel Buildings, Including Supplement No. 1, dated November 16, 2005.

AISC 358-05, Prequalified Connections For Special And Intermediate Steel Moment Frames for Seismic Applications, dated December 13, 2005, including Supplement No. 1, dated June 18, 2009.

AISC 360-05, Specification For Structural Steel Buildings, dated March 9, 2005.

AISC Design Guide 4, Second Edition, Extended End-Plate Moment connections, Seismic And Wind Applications, 2004.

AISC Design Guide 16, Flush And Extended Multiple-Row Moment End Connections, 2003.

ASCE 7-05, Minimum Design Loads For Buildings And Other Structures, 2005.

Fluor Guideline, 000.215.1216, Earthquake Engineering, latest edition.

Fluor Standard Drawing, 000.215.5040, Structural Steel Connection Details, latest edition.

Fluor Standard Drawing, 000.215.5065, Sheet 1, Structural Steel Seismic Connection Details, latest edition.

ATTACHMENTS

Attachment 01	Sample Design 1 – Bolted End-Plate Moment Connection With Seismic Detailing (30Oct09)

SAMPLE DESIGN 1: BOLTED END-PLATE MOMENT CONNECTION WITH SEISMIC DETAILING

GIVEN:

Operating Weights (including pipes and steel):

Flare Header: 2 K

Upper Deck: 48 K

Lower Deck: 75 K

Seismic Data:

Site Class $= E$

Occupancy Category $= III$

$I = 1.25$

$S_s = 1.80$

$S_1 = 0.60$

$T_L = 10\,\text{sec}$

$T = 0.65\,\text{sec}$

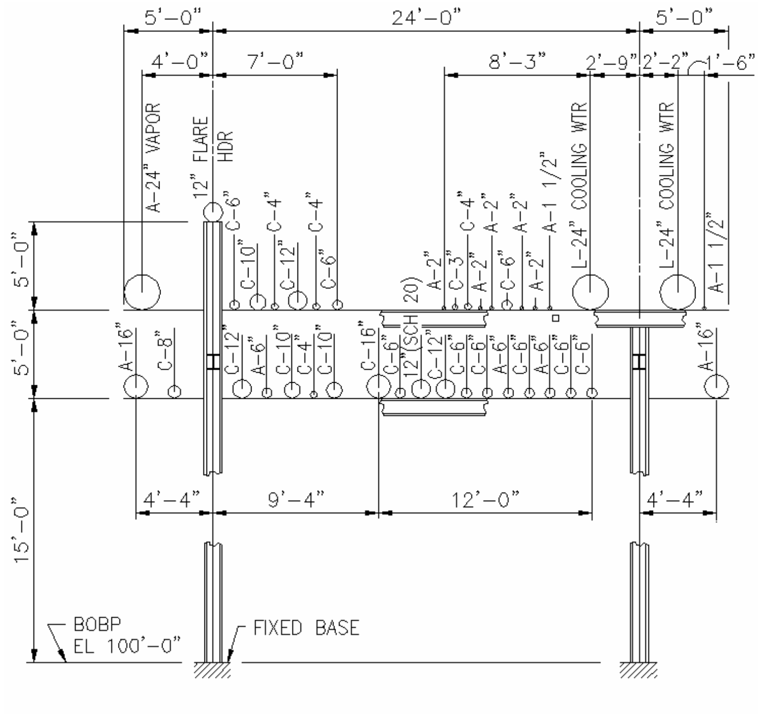
Intermediate Moment Frame (IMF)

Pipe Support Spacing: 20'-0"

Lower Deck Member Sizes:

Beam – W14x48

Column – W14x99



PIPE SUPPORT

REQUIRED:

Design a Four Bolt Unstiffened (4E) bolted end plate moment connection for the lower deck IMF of the piperack.

SOLUTION:

See proceeding calculations spreadsheet.

Design Input:

Connection Dimensions & Properties:

$b_p := 9\text{in}$

$g := 5.5\text{in}$

$d_{bolt} := 1.5\text{in}$

$A_{bolt} = 1.77 \cdot \text{in}^2$

$p_{fo} := 2\text{in}$

$p_{fi} := 2\text{in}$

$s := 3.71\text{in}$

$h_1 := 10.2\text{in}$

$h_0 := 13.68\text{in}$

$d_e := 2.75\text{in}$

$t_{plate} := 1.25\text{in}$

$D_{pl} := \frac{1}{4}\text{in}$

$t_s := 0.75\text{in}$

$D_{stf} := \frac{5}{16}\text{in}$

$F_y := 50 \cdot \text{ksi}$

$F_u := 65\text{ksi}$

$E_s := 29000 \cdot \text{ksi}$

Beam Shear Force (lower deck):

$$V_{gravity} := \frac{1.2 \cdot 75\text{kips}}{2}$$

$V_{gravity} = 45.00 \cdot \text{kips}$

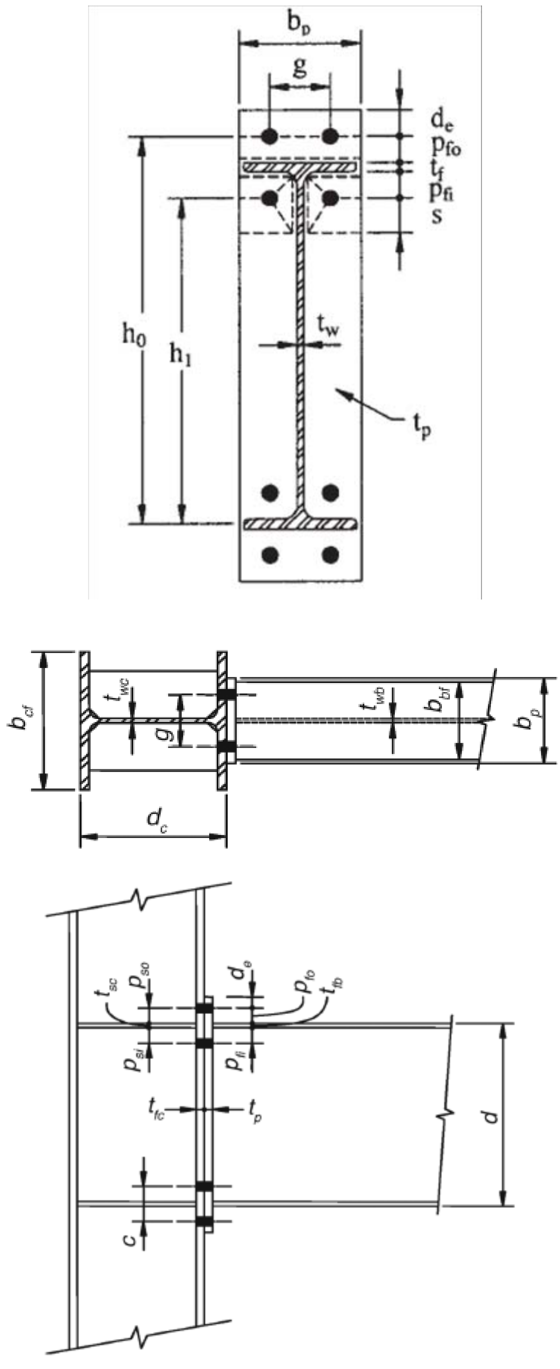


Fig. 6.2. Four-bolt unstiffened extended (4E) end-plate geometry.

Reference: AISC 341-05, AISC 358-05

Beam Section Properties:

Member := "w14x48"

Extracted Properties

$A = 14.10 \cdot in^2$	$A_b := A$
$d = 13.79 \cdot in$	$d_b := d$
$t_w = 0.34 \cdot in$	$t_{wb} := t_w$
$b_f = 8.03 \cdot in$	$b_{fb} := b_f$
$t_f = 0.60 \cdot in$	$t_{fb} := t_f$
$k = 1.38 \cdot in$	$k_b := k$
$Z_x = 78.40 \cdot in^3$	$Z_{xb} := Z_x$
$Z_y = 19.60 \cdot in^3$	$Z_{yb} := Z_y$
$r_x = 5.85 \cdot in$	$r_{xb} := r_x$
$r_y = 1.91 \cdot in$	$r_{yb} := r_y$
$g_b := 5.5in$	Working gage

Column Section Properties:

Member := "w14x99"

Extracted Properties

$A = 29.10 \cdot in^2$	$A_c := A$
$d = 14.16 \cdot in$	$d_c := d$
$t_w = 0.485 \cdot in$	$t_{wc} := t_w$
$b_f = 14.56 \cdot in$	$b_{fc} := b_f$
$t_f = 0.78 \cdot in$	$t_{fc} := t_f$
$k = 1.44 \cdot in$	$k_c := k$
$Z_x = 173.00 \cdot in^3$	$Z_{xc} := Z_x$
$Z_y = 83.60 \cdot in^3$	$Z_{yc} := Z_y$
$r_x = 6.17 \cdot in$	$r_{xc} := r_x$
$r_y = 3.71 \cdot in$	$r_{yc} := r_y$
$g_c := 5.5in$	Working gage

$\phi_d := 1.0$ $\phi_n := 0.9$ **Resistance Factors: [AISC 358-05 Sect. 2.4.1]**

$l_{bm} := 24ft$ **Beam Length CL-CL**

$Weld_{strial} := \frac{5}{16} in$ **Stiffener Weld Size (Trial)**

Beam Side Design

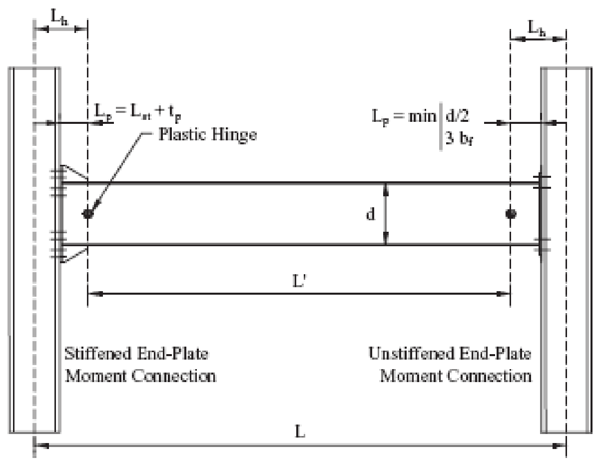


Fig. 2.1. Location of plastic hinges.

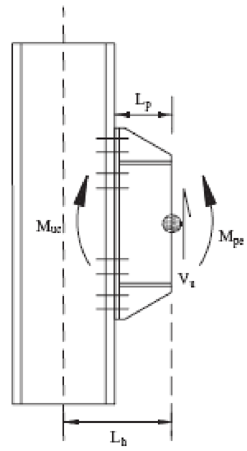


Fig. 2.2. Calculation of connection design moment.

$L_p = S_h$

$L := l_{bm} - d_c$ $\quad L = 22.82\ ft$ **Beam Length (within column flange):**

1. Connection Design Moment

$R_y := 1.1$ **[AISC 341-05 Table I-6-1]**

$C_{pr} := \min\left[\left(\frac{F_y + F_u}{2 \cdot F_y}\right), 1.2\right]$ $\quad C_{pr} = 1.15$ **[AISC 358-05 (eqn. 2.4.3-2)]**

$M_{pe} := C_{pr} \cdot R_y \cdot F_y \cdot Z_{xb}$ $\quad M_{pe} = 4958.80 \cdot kip \cdot in$ **[AISC 358-05 (eqn. 6.9-3)]**

Location of Plastic Hinge

$S_h := \min\left(\frac{d_b}{2}, 3 \cdot b_{fb}\right)$ $\quad S_h = 6.89 \cdot in$ **[AISC 358-05 (eqn. 6.9-4)]**

Distance between Plastic Hinge

$L_{prime} := L - 2 \cdot \left(S_h + t_{plate}\right)$ $\quad L_{prime} = 21.46\ ft$

Shear Capacity plus Shear Gravity

$V_u := 2 \cdot \frac{M_{pe}}{L_{prime}} + V_{gravity}$ $\quad V_u = 83.51 \cdot kips$ **[AISC 358-05 (eqn. 6.9-16)]**

Moment at the Face of Column (Connection Design Moment)

$M_f := M_{pe} + V_u \cdot S_h$ $\quad M_f = 5534.58 \cdot kip \cdot in$ **[AISC 358-05 (eqn. 6.9-2)]**

2. Select Connection Configuration: Four-Bolt Extended Unstiffened

Refer to Detail 1 on Standard Drawing 000.215.5065 for Geometric Design Data

$b_p := \mathrm{Ceil}(b_{fb} + 1in, 0.25in) = 9.25 \cdot in$

$g := g_b$

$p_{fi} := 3in - t_{fc}$ $\quad p_{fi} = 2.22 \cdot in$

$p_{fo} := 2in$

$d_e := \mathrm{Ceil}(1.75 \cdot d_{bolt}, 0.25in) = 2.75 \cdot in$ **[AISC 360 Table J3.4]**

$F_{yp} := 50ksi \quad F_{up} := 65ksi$ ASTM A992 Gr. 50 Steel

$F_{nt} := 90ksi$ ASTM A325 Bolts

$h_0 := d_b + p_{fo} - \frac{t_{fb}}{2}$ $\qquad h_0 = 15.49 \cdot in$

$h_1 := d_b + t_{fb} - p_{fo} - \frac{t_{fb}}{2}$ $\qquad h_1 = 12.09 \cdot in$

3. Determine the Required Bolt Diameter (ASTM A325)

$$d_{breqd} := \sqrt{\frac{2M_f}{\pi \cdot \phi_d \cdot F_{nt} \cdot (h_0 + h_1)}}$$ **[AISC 358-05 (eqn. 6.9-6)]**

$d_{breqd} = 1.19 \cdot in$

4. Check Bolt Diameter and Calculate the No Prying Bolt Moment

$$d_{bolt} := \begin{cases} d_{bolt} & \text{if } d_{breqd} < d_{bolt} \\ \text{"Change d.bolt"} & \text{otherwise} \end{cases} = 1.50 \cdot in$$

$\boxed{d_{bolt} = 1.50 \cdot in}$ ASTM A325

5. Determine the Required End Plate Thickness End Plate Yield Line Mechanism Parameter

AISC 358 Table 6.2

$s := \frac{1}{2} \cdot \sqrt{b_p \cdot g}$ $\qquad s = 3.57 \cdot in$

$s := \max(p_{fi}, s)$

$$Y_{p1} := \frac{b_p}{2} \cdot \left[h_1 \cdot \left(\frac{1}{p_{fi}} + \frac{1}{s}\right) + h_0 \cdot \left(\frac{1}{p_{fo}}\right) - \frac{1}{2}\right] + \frac{2}{g} \cdot [h_1 \cdot (p_{fi} + s)] \qquad Y_{p1} = 99.81 \cdot in$$

Required End Plate Thickness

$$t_{preqd} := \sqrt{1.11 \cdot \frac{M_f}{\phi_d \cdot F_{yp} \cdot Y_{p1}}} \qquad t_{preqd} = 1.11 \cdot in$$ **[AISC 358-05 (eqn. 6.9-8)]**

6. Check End Plate Thickness

$$\text{check1} := \begin{cases} \text{"OK"} & \text{if } t_{preqd} < t_{plate} \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$\text{check1} = \text{"OK"}$ ASTM A572 Gr. 50 steel

7. Calculate the Factored Beam Flange Force

$$F_{fu} := \frac{M_f}{(d_b - t_{fb})} \qquad F_{fu} = 419.45 \cdot \text{kips}$$

[AISC 358-05 (eqn. 6.9-9)]

8. Check Shear Yielding of Extended Portion of End Plate

$$\phi R_{n1} := \phi_d \cdot 0.6 \cdot F_{yp} \cdot b_p \cdot t_{plate} \qquad \phi R_{n1} = 346.87 \cdot \text{kips}$$

[AISC 358-05 (eqn. 6.9-10)]

Check Inequality 3.12

$$\text{check2} := \begin{cases} \text{"OK"} & \text{if } \frac{F_{fu}}{2} < \phi R_{n1} \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$\text{check2} = \text{"OK"}$

9. Check Shear Rupture of Extended Portion of End Plate

$$A_n := \left[b_p - 2 \cdot \left(d_{bolt} + \frac{1}{8} \text{in} \right) \right] \cdot t_{plate} \qquad A_n = 7.50 \cdot \text{in}^2$$

[AISC 358-05 (eqn. 6.9-12)]

$$\phi R_{n2} := \phi_n \cdot (0.6 \cdot F_{up}) \cdot A_n \qquad \phi R_{n2} = 263.25 \cdot \text{kips}$$

[AISC 358-05 (eqn. 6.9-11)]

Check Inequality 3.13

$$\text{check3} := \begin{cases} \text{"OK"} & \text{if } \frac{F_{fu}}{2} < \phi R_{n2} \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$\text{check3} = \text{"OK"}$

10. Unstiffened connection ===> This step not required

11. Check Compression Bolts Shear Rupture Strength

$V_u = 83.51 \cdot \text{kips}$

$n_b := 4$

$F_v := 60\text{ksi}$

$$\phi R_{n3} := \phi_n \cdot n_b \cdot F_v \cdot A_{bolt} \qquad \phi R_{n3} = 382.32 \cdot \text{kips}$$

[AISC 358-05 (eqn. 6.9-15)]

$$\text{check4} := \begin{cases} \text{"OK"} & \text{if } V_u < \phi R_{n3} \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$\text{check4} = \text{"OK"}$

12. Check Compression Bolts Bearing/Tearout

i) End Plate

$n_i := 2 \quad n_o := 2$

$L_{ci} := p_{fi} + t_{fb} + p_{fo} - \left(d_{bolt} + \frac{1}{16} in\right)$ $\quad L_{ci} = 3.25 \cdot in$

$L_{co} := d_e - \left(d_{bolt} + \frac{1}{16} in\right)$ $\quad L_{co} = 1.19 \cdot in$

$r_i := \min\left(1.2 \cdot L_{ci} \cdot t_{plate} \cdot F_{up}, 2.4 \cdot d_{bolt} \cdot t_{plate} \cdot F_{up}\right)$ [AISC 358-05 (eqn. 6.9-18)]

$r_i = 292.50 \cdot kips$

$r_o := \min\left(1.2 \cdot L_{co} \cdot t_{plate} \cdot F_{up}, 2.4 \cdot d_{bolt} \cdot t_{plate} \cdot F_{up}\right)$ [AISC 358-05 (eqn. 6.9-19)]

$r_o = 115.78 \cdot kips$

$\phi R_{n4} := \phi_n \cdot n_i \cdot r_i + \phi_n \cdot n_o \cdot r_o$ $\quad \phi R_{n4} = 734.91 \cdot kips$ [AISC 358-05 (eqn. 6.9-17)]

$check5 := \begin{cases} \text{"OK"} & \text{if } V_u < \phi R_{n4} \\ \text{"NG"} & \text{otherwise} \end{cases}$ $\quad \boxed{check5 = \text{"OK"}}$

ii) Column Flange

$t_{fc} = 0.78 \cdot in$

$\phi R_{n5} := \phi R_{n4} \cdot \left(\frac{t_{fc}}{t_{plate}}\right) \cdot \left(\frac{F_{yp}}{F_y}\right)$ $\quad \phi R_{n5} = 458.58 \cdot kips$

$check6 := \begin{cases} \text{"OK"} & \text{if } V_u < \phi R_{n5} \\ \text{"NG"} & \text{otherwise} \end{cases}$ $\quad \boxed{check6 = \text{"OK"}}$

13. Design Welds

i) Beam Flanges and End-Plate Weld

Use CJP Welds

ii) Beam Web to End-Plate Weld

Required weld to develop the bending stress in the beam web near the tension bolts using E70

$D := 0.6 \cdot \frac{F_y \cdot t_{wb}}{2 \cdot 1.392 \frac{kip}{in}}$ $\quad D = 3.66$

$$\mathbf{Weld} := \frac{\mathbf{in} \cdot \mathbf{ceil}(\mathbf{D})}{16} \qquad \boxed{\mathbf{Weld} = \frac{1}{4} \cdot \mathbf{in}}$$

Effective length of weld

$$\mathbf{l_{weld}} := \frac{\mathbf{d_b}}{2} - \mathbf{t_{fb}} \qquad \mathbf{l_{weld}} = 6.30 \cdot \mathbf{in}$$

$$\mathbf{D} := \frac{\mathbf{V_u}}{2 \cdot 1.392 \frac{\mathbf{kip}}{\mathbf{in}} \cdot \mathbf{l_{weld}}} \qquad \mathbf{D} = 4.76$$

$$\mathbf{Weld} := \frac{\mathbf{in} \cdot \mathbf{ceil}(\mathbf{D})}{16} \qquad \boxed{\mathbf{Weld} = \frac{5}{16} \cdot \mathbf{in}}$$

Column-Side Design

14a. AISC 341 Sec. 11.5 for Continuity Plates

$$\mathbf{t_{fcmin1}} := 0.54 \cdot \sqrt{\mathbf{b_{fc}} \cdot \mathbf{t_{fb}} \cdot \frac{\mathbf{F_y}}{\mathbf{F_y}}} \qquad \mathbf{t_{fcmin1}} = 1.59 \cdot \mathbf{in}$$

$$\mathbf{t_{fcmin2}} := \frac{\mathbf{b_{fc}}}{6} \qquad \mathbf{t_{fcmin2}} = 2.43 \cdot \mathbf{in}$$

$$\mathbf{check7} := \begin{cases} \text{"OK"} & \textbf{if } \mathbf{t_{fc}} > \mathbf{max}(\mathbf{t_{fcmin1}}, \mathbf{t_{fcmin2}}) \\ \text{"Add Flange Stiffners"} & \textbf{otherwise} \end{cases} \qquad \boxed{\mathbf{check7} = \text{"Add Flange Stiffners"}}$$

14b. AISC 360 Sec. J10 Requirements

J10.1: Column Flange Local Bending Strength

$$\phi \mathbf{R_{n6a}} := 6.25 \cdot \mathbf{t_{fc}}^2 \cdot \mathbf{F_y} \qquad \phi \mathbf{R_{n6a}} = 190.13 \cdot \mathbf{kips}$$

J10.8: Additional Requirements for Stiffeners

$$\mathbf{t_{stmin}} := 0.5 \cdot \mathbf{t_{fb}} \qquad \mathbf{t_{stmin}} = 0.30 \cdot \mathbf{in}$$

Minimum Stiffener thickness

$$\mathbf{t_{stmax}} := \frac{\mathbf{b_{fc}}}{15} \qquad \mathbf{t_{stmax}} = 0.97 \cdot \mathbf{in}$$

Maximum Stiffener thickness

14c. Check the Column Flange for Flexural Yielding

AISC 358-05 Table 6.2 eqns.:

$$\mathbf{s} := \frac{1}{2} \cdot \sqrt{\mathbf{b_{fc}} \cdot \mathbf{g}} \qquad \mathbf{s} = 3.57 \cdot \mathbf{in}$$

$$\mathbf{c} := \mathbf{p_{fo}} + \mathbf{t_{fb}} + \mathbf{p_{fi}} \qquad \mathbf{c} = 1.18 \times 10^{10} \, \mathbf{s}^{-1.00} \cdot \mathbf{in}$$

$$Y_{c2} := \frac{b_{fc}}{2}\left[h_1\cdot\left(\frac{1}{s}\right) + h_0\cdot\left(\frac{1}{s}\right)\right] + \frac{2}{g}\left[h_1\cdot\left(s + \frac{3\cdot c}{4}\right) + h_0\cdot\left(s + \frac{c}{4}\right) + \frac{c^2}{2}\right] + \frac{g}{2}$$

$$Y_{c2} = 119.38\cdot in$$

Required Unstiffened Column Flange Thickness

$t_{fcreqd} := \sqrt{1.11\cdot\frac{M_f}{\phi_d\cdot F_y\cdot Y_{c2}}}$ $t_{fcreqd} = 1.01\cdot in$ **[AISC 358-05 (eqn. 6.9-20)]**

$$check8 := \begin{cases} \text{"OK"} & \textbf{if } t_{fcreqd} < t_{fc} \\ \text{"Add Flange Stiffners"} & \textbf{otherwise} \end{cases}$$

$check8 = \text{"Add Flange Stiffners"}$

AISC 358-05 Table 6.5 eqns.:

$p_{so} := \frac{c - t_s}{2}$ $p_{so} = 2.03\cdot in$ $p_{si} := p_{so}$

$$Y_{c3} := \frac{b_{fc}}{2}\left[h_1\cdot\left(\frac{1}{s} + \frac{1}{p_{si}}\right) + h_0\cdot\left(\frac{1}{s} + \frac{1}{p_{so}}\right)\right] + \frac{2}{g}\cdot\left[h_1\cdot(s + p_{si}) + h_0\cdot(s + p_{so})\right]$$

$$Y_{c3} = 208.97\cdot in$$

$t_{fcreqd} := \sqrt{1.11\cdot\frac{M_f}{\phi_d\cdot F_y\cdot Y_{c3}}}$ $t_{fcreqd} = 0.7668\cdot in$ **[AISC 358-05 (eqn. 6.9-20)]**

$$check8a := \begin{cases} \text{"OK"} & \textbf{if } t_{fcreqd} \le t_{fc} \\ \text{"Column Flange NG"} & \textbf{otherwise} \end{cases}$$

$check8a = \text{"OK"}$

$$check8b := \begin{cases} \text{"OK"} & \textbf{if } t_{stmin} < t_s < t_{stmax} \\ \text{"Stiffner thickness NG"} & \textbf{otherwise} \end{cases}$$

$check8b = \text{"OK"}$

15. Calculate Strength of Unstiffened Column Flange to Determine Stiffener Design Force

$\phi M_{cf} := \phi_d\cdot F_y\cdot Y_{c2}\cdot t_{fc}^{\,2}$ $\phi M_{cf} = 3631.63\cdot kip\cdot in$ **[AISC 358-05 (eqn. 6.9-21)]**

$\phi R_{n6} := \frac{\phi M_{cf}}{d_b - t_{fb}}$ $\phi R_{n6} = 275.23\cdot kips$ **[AISC 358-05 (eqn. 6.9-22)]**

$$check9 := \begin{cases} \text{"OK"} & \textbf{if } F_{fu} > \phi R_{n6} \\ \text{"NG"} & \textbf{otherwise} \end{cases}$$

$check9 = \text{"OK"}$

16. Calculate Local Web Yielding Strength

Top of beam flange is well below column top $C_t := 1.0$

$$\phi R_{n7} := \phi_d \cdot C_t \cdot \left(6k_c + t_{fb} + 2t_{plate}\right) \cdot F_y \cdot t_{wc} \qquad \phi R_{n7} = 284.21 \cdot kips$$ **[AISC 358-05 (eqn. 6.9-24)]**

$$check10 := \begin{cases} \text{"OK"} & \text{if } F_{fu} > \phi R_{n7} \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad \boxed{check10 = \text{"OK"}}$$

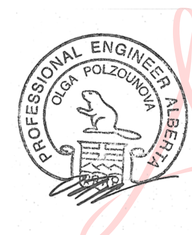
17. Calculate Web Buckling Strength

$ht_{wc} := 21.7 \qquad \phi := 0.75$

$h := ht_{wc} \cdot t_{wc} \qquad h = 10.52 \cdot in$

$$\phi R_{n8} := \phi \cdot 24 \cdot t_{wc}{}^3 \cdot \frac{\sqrt{E_s \cdot F_y}}{h} \qquad \phi R_{n8} = 234.95 \cdot kips$$ **[AISC 358-05 (eqn. 6.9-26)]**

$$check11 := \begin{cases} \text{"OK"} & \text{if } F_{fu} > \phi R_{n8} \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad \boxed{check11 = \text{"OK"}}$$



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18. Calculate Web Crippling Strength

Assumes Column is at top (AISC 360 J10.3)

$N := t_{fb} + 0 = 0.60 \cdot in$

$$\phi R_{n9} := \phi \cdot 0.4 \cdot t_{wc}{}^2 \cdot \left[1 + 3 \cdot \left(\frac{N}{d_c}\right) \cdot \left(\frac{t_{wc}}{t_{fc}}\right)^{1.5}\right] \cdot \sqrt{E_s \cdot \frac{F_y \cdot t_{fc}}{t_{wc}}} = 114.42 \cdot kips$$ **AISC 358-05 (eqn. 6.9-29)**

$$check12 := \begin{cases} \text{"OK"} & \text{if } F_{fu} > \phi R_{n9} \\ \text{"NG"} & \text{otherwise} \end{cases} \qquad \boxed{check12 = \text{"OK"}}$$

19. Determine Stiffener Design Force

$$\phi R_{nstiff} := \min\left(\phi R_{n6a}, \phi R_{n6}, \phi R_{n7}, \phi R_{n8}, \phi R_{n9}\right) = 114.42 \cdot kips$$ **AISC 358-05 (eqn. 6.9-32)**

$$F_{cu} := F_{fu} - \phi R_{nstiff} \qquad F_{cu} = 305.02 \cdot kips$$

20. Stiffener Design Strength

Gross Area of Stiffeners at both sides of the web

$$A_g := t_s \cdot \left(b_{fc} - t_{wc}\right) \qquad A_g = 10.56 \cdot in^2$$

$\phi_t := 0.9$ **Tensile Strength (AISC Chapter D)**

$$\phi P_{nt} := \phi_t \cdot F_y \cdot A_g \qquad \phi P_{nt} = 475.20 \cdot kips$$

$$check13 := \begin{cases} \text{"OK"} & \text{if } \phi P_{nt} > F_{cu} \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$check13 = \text{"OK"}$

Compressive Strength (AISC 360 Section E6.2)

$\phi_c := 0.9$

$L := h \qquad r := \dfrac{t_s}{\sqrt{12}} \qquad K := 0.5 \qquad \dfrac{K \cdot L}{r} = 24.31 \qquad 4.71 \cdot \sqrt{\dfrac{E_s}{F_y}} = 113.43$

$$F_e := \frac{\pi^2 \cdot E_s}{\left(\dfrac{K \cdot L}{r}\right)^2} \qquad F_e = 484.50 \cdot ksi$$

$$F_{cr} := 0.658^{\frac{F_y}{F_e}} \cdot F_y \qquad F_{cr} = 47.89 \cdot ksi$$

$$\phi P_{nc} := \phi_c \cdot F_{cr} \cdot A_g \qquad \phi P_{nc} = 455.11 \cdot kips$$

$$check14 := \begin{cases} \text{"OK"} & \text{if } \phi P_{nc} > F_{cu} \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$check14 = \text{"OK"}$

21. Stiffener to Column Weld Design

Required weld to resist differences between required strength and available limit state strength using E70

i) Flange Welds

$D := Weld_{strial} \cdot 16 \qquad l := 4(b_{fc} - 0.75in)$

$$\phi R_{wt} := 0.75 \cdot 0.6 \cdot 70ksi \cdot \frac{\sqrt{2}}{2} \cdot \frac{D}{16} \cdot l \qquad \phi R_{wt} = 384.64 \cdot kips$$

$$check15 := \begin{cases} \text{"OK"} & \text{if } \phi R_{wt} > F_{cu} \\ \text{"NG"} & \text{otherwise} \end{cases}$$

$check15 = \text{"OK"}$

$$Weld := \begin{cases} Weld_{strial} & \text{if } \phi R_{wt} > F_{cu} \\ \text{"Change Weld Size"} & \text{otherwise} \end{cases}$$

$Weld = \dfrac{5}{16} \cdot in$

ii) Web Welds : Not Required for full-depth stiffeners, however, check difference between required strength and web buckling strength

$\mathbf{D} := \mathbf{Weld_{strial}} \cdot 16$ $\qquad$ $\mathbf{l} := 4 \cdot \mathbf{h}$

$\phi\mathbf{R_{wt}} := 0.75 \cdot 0.6 \cdot 70\mathbf{ksi} \cdot \frac{\sqrt{2}}{2} \cdot \frac{\mathbf{D}}{16} \cdot \mathbf{l}$ $\qquad$ $\phi\mathbf{R_{wt}} = 293.03 \cdot \mathbf{kips}$

$\mathbf{F_{reqd}} := \mathbf{F_{cu}} - \phi\mathbf{R_{n9}}$ $\qquad$ $\mathbf{F_{reqd}} = 190.60 \cdot \mathbf{kips}$

$\mathbf{check16} := \begin{vmatrix} \text{"OK"} & \textbf{if} \quad \phi\mathbf{R_{wt}} > \mathbf{F_{reqd}} \\ \text{"NG"} & \textbf{otherwise} \end{vmatrix}$ $\qquad$ $\boxed{\mathbf{check16} = \text{"OK"}}$

$\mathbf{Weld} := \begin{vmatrix} \mathbf{Weld_{strial}} & \textbf{if} \quad \phi\mathbf{R_{wt}} > \mathbf{F_{reqd}} \\ \text{"Change Weld Size"} & \textbf{otherwise} \end{vmatrix}$ $\qquad$ $\boxed{\mathbf{Weld} = \frac{5}{16} \cdot \mathbf{in}}$